PLA Year 3, Semester A: mini-project questions

Each student is allocated two questions, drawn at random from the questions below for the Year 2 modules they have taken. Solutions must follow the guidelines and are marked according to the mark scheme in *Mini-project and viva: guidelines and mark scheme*.

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Question MTH5112–1 (Bases inside a spanning set)**.** Let V be a vector space and let $S \subseteq V$ be a finite set that spans V.

(a) Prove that, for a subset $B \subseteq S$, the following are equivalent: (i) B is a basis of V; (ii) B spans V, but no proper subset of B does; (iii) B is linearly independent, but $B \cup \{\mathbf{s}\}$ is linearly dependent for every $\mathbf{s} \in S \setminus B$. Deduce that all such subsets B have the same number of elements.

(b) Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_5\} \subseteq \mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{pmatrix}1\\0\\1\end{pmatrix},\ \mathbf{v}_2 = \begin{pmatrix}0\\1\\1\end{pmatrix},\ \mathbf{v}_3 = \begin{pmatrix}1\\1\\2\end{pmatrix},\ \mathbf{v}_4 = \begin{pmatrix}1\\-1\\0\end{pmatrix},\ \mathbf{v}_5 = \begin{pmatrix}0\\0\\1\end{pmatrix}.$$

Find all subsets of S that are bases of $\mathbb{R}^3$, and explain how your answer illustrates (a).

Question MTH5112–2 (Extending an independent set to a basis)**.** Let V be a finite-dimensional vector space.

(a) Let $L \subseteq S \subseteq V$, where L is linearly independent and S is a finite set spanning V. Prove that there is a basis B of V with $L \subseteq B \subseteq S$, and that every such basis has exactly $\dim V$ elements. Is B uniquely determined by L and S?

(b) In $V = P_2$, let $L = \{1 + x^2,\ x\}$ and

$$S = L \cup \{\,1,\ \ x + x^2,\ \ 2 + x + 2x^2,\ \ 1 - x^2\,\}.$$

Verify that S spans P_2, and find all bases B of P_2 with $L \subseteq B \subseteq S$.

Question MTH5112–3 (Translating a basis)**.** Let V be a vector space, let $\mathbf{v}_1, \ldots, \mathbf{v}_n \in V$ be linearly independent, and let $\mathbf{w} \in V$.

(a) Suppose that $\mathbf{v}_1 + \mathbf{w}, \ldots, \mathbf{v}_n + \mathbf{w}$ are linearly dependent. Prove that $\mathbf{w} \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

(b) Suppose now that $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis of V and $\mathbf{w} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Prove that $\{\mathbf{v}_1 + \mathbf{w}, \ldots, \mathbf{v}_n + \mathbf{w}\}$ is a basis of V if and only if $a_1 + \cdots + a_n \neq -1$, and that otherwise $\operatorname{span}(\mathbf{v}_1 + \mathbf{w}, \ldots, \mathbf{v}_n + \mathbf{w})$ has dimension $n - 1$.

Question MTH5112–4 (Exchanging basis vectors)**.** Let V be a vector space with a basis $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$.

(a) Let $\mathbf{w} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n \in V$ and fix $k \in \{1, \ldots, n\}$. Prove that the set obtained from B by replacing $\mathbf{v}_k$ with $\mathbf{w}$ is a basis of V if and only if $a_k \neq 0$.

(b) Let C be another basis of V. Prove that for every $\mathbf{b} \in B$ there exists $\mathbf{c} \in C$ such that $(B \setminus \{\mathbf{b}\}) \cup \{\mathbf{c}\}$ is a basis of V. Find all such $\mathbf{c}$ when $V = \mathbb{R}^3$, $\mathbf{b} = \mathbf{e}_1$, and

$$B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}, \qquad C = \{(1,1,0)^T,\ (0,1,1)^T,\ (1,0,1)^T\}.$$

Question MTH5212–1 (Bases and coordinates)**.** For vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^n$, let P be the $n \times n$ matrix with columns $\mathbf{b}_1, \ldots, \mathbf{b}_n$.

(a) Prove that $\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis of $\mathbb{R}^n$ if and only if P is invertible, and that in this case the coordinate vector of $\mathbf{v} \in \mathbb{R}^n$ with respect to the ordered basis $B = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$ is $[\mathbf{v}]_B = P^{-1}\mathbf{v}$.

(b) For $a \in \mathbb{R}$, consider

$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{b}_4 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ a \end{pmatrix}.$$

Find all values of a for which $B = (\mathbf{b}_1, \ldots, \mathbf{b}_4)$ is a basis of $\mathbb{R}^4$. For each value of a for which it is not, find a non-trivial linear relation between $\mathbf{b}_1, \ldots, \mathbf{b}_4$ and explain why $\mathbf{v} = (2, 3, 3, 2)^T$ then has no well-defined coordinate vector. For $a = 2$, find $[\mathbf{v}]_B$.

Question MTH5212–2 (Rank–Nullity and linear systems)**.** Let A be an $m \times n$ real matrix.

(a) Prove that if $\mathbf{x}_p$ is a solution of $A\mathbf{x} = \mathbf{b}$, then the set of all solutions is $\{\mathbf{x}_p + \mathbf{z} : \mathbf{z} \in \ker A\}$. Deduce, using the Rank–Nullity Theorem, that every consistent system $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if $\operatorname{rank} A = n$.

(b) Let

$$A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 1 \\ -1 & -2 & 2 & 4 \\ 3 & 6 & 1 & 7 \end{pmatrix}.$$

Find the rank of A and a basis of $\ker A$, and verify the Rank–Nullity Theorem for A. By row reducing the augmented matrix $(A \,|\, \mathbf{b})$, find a single linear equation in $b_1, \ldots, b_4$ that is satisfied if and only if $A\mathbf{x} = \mathbf{b}$ is consistent, and find the general solution of $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = (3, 4, 5, 11)^T$.

Question MTH5212–3 (Diagonalisation and linear recurrences)**.** (a) Prove that eigenvectors of a square matrix corresponding to three distinct eigenvalues are linearly independent, and deduce that a 3×3 matrix with three distinct eigenvalues is diagonalisable. Prove also that if $A = PDP^{-1}$, then $A^k = PD^kP^{-1}$ for every integer $k \geq 1$.

(b) Let

$$A = \begin{pmatrix} 2 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Diagonalise A, i.e. find an invertible P and a diagonal D with $A = PDP^{-1}$ (you need not compute P^{-1}). Hence solve the recurrence $\mathbf{x}_{k+1} = A\mathbf{x}_k$ with $\mathbf{x}_0 = (1, 1, 1)^T$, and determine all initial vectors $\mathbf{x}_0$ for which the sequence $(\mathbf{x}_k)$ is bounded.

Question MTH5212–4 (Least squares)**.** Let A be an $m \times n$ real matrix and $\mathbf{b} \in \mathbb{R}^m$.

(a) Prove that if $A^TA\mathbf{x}^* = A^T\mathbf{b}$, then $\mathbf{x}^*$ minimises $\|A\mathbf{x} - \mathbf{b}\|$ over all $\mathbf{x} \in \mathbb{R}^n$, and that if the columns of A are linearly independent, then A^TA is invertible, so the normal equations have a unique solution.

(b) Find the straight line $y = c_0 + c_1 t$ that best fits the data points $(t, y) = (1, 2)$, $(2, 2)$, $(3, 4)$ in the least squares sense. Compute the residual vector, verify that it is orthogonal to the columns of the relevant matrix, and find the least squares error. Explain why the best-fitting polynomial of degree at most 2 fits these data exactly.

Question MTH5104–1 (Suprema and maxima)**.** Let $A \subseteq \mathbb{R}$ be non-empty and bounded above.

(a) Prove that an upper bound s of A is the supremum of A if and only if for every $\varepsilon > 0$ there exists $a \in A$ with $a > s - \varepsilon$. Deduce that there is a sequence of elements of A converging to $\sup A$, and that A has a maximum if and only if $\sup A \in A$.

(b) Let

$$A = \{z \in \mathbb{R} : z^2 < 11\}, \qquad B = \left\{\frac{1}{m} - \frac{1}{n} : m, n \in \mathbb{N}\right\}.$$

For each of A and B, find the supremum and the infimum, and decide whether the set has a maximum and a minimum.

Question MTH5104–2 (Limits of sequences)**.** (a) Prove that a convergent sequence of real numbers is bounded, and that its limit is unique. Prove that if $x_n \to L$ and $y_n \to M$, then $x_n y_n \to LM$.

(b) Let

$$x_n = \frac{(-1)^n}{n^2} + \frac{3n-4}{n}, \qquad y_n = \frac{(-1)^n\, n}{n+1}.$$

Prove *directly from the definition* that (x_n) converges, giving an explicit N for each $\varepsilon > 0$. Prove directly from the definition that (y_n) does not converge, but that (y_n^2) does, and explain why this does not contradict (a).

Question MTH5104–3 (Series and the ratio test)**.** Let $(a_k)_{k\geq 1}$ be a sequence of positive real numbers such that $a_{k+1}/a_k \to \ell$.

(a) Prove that if $\ell < 1$ then $\sum_{k=1}^{\infty} a_k$ converges, and that if $\ell > 1$ then it diverges. Show by examples that no conclusion can be drawn when $\ell = 1$.

(b) Determine all $x > 0$ for which the series

$$\sum_{k=1}^{\infty} \frac{(3^k + 5^k)\, x^k}{2^k + 7^k}$$

converges.

Question MTH5104–4 (Continuity)**.** (a) Prove that a function $f\colon \mathbb{R} \to \mathbb{R}$ is continuous at $a \in \mathbb{R}$ if and only if $f(x_n) \to f(a)$ for every sequence (x_n) with $x_n \to a$.

(b) For $c \in \mathbb{R}$, let

$$f_c(x) = \begin{cases} 2x + c & \text{if } x \geq 0, \\ 3x & \text{if } x < 0, \end{cases} \qquad g(x) = \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Find all c for which f_c is continuous at 0, and find all points at which g is continuous.

Question MTH5115–1 (Feasible solutions that are not basic)**.** Consider a linear program in standard equation form, maximise $\mathbf{c}^T\mathbf{x}$ subject to $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$, where A is a real $m \times 4$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{c} \in \mathbb{R}^4$. Let $\mathbf{x} = (x_1, x_2, x_3, 0)^T$ be a feasible solution with $x_1, x_2, x_3 > 0$.

(a) Suppose that $\mathbf{x}$ is not basic feasible. Prove that there is a non-zero vector $\mathbf{d} = (d_1, d_2, d_3, 0)^T$ satisfying $A\mathbf{d} = \mathbf{0}$ and a number $\theta > 0$ such that $\mathbf{x} + \theta\mathbf{d}$ and $\mathbf{x} - \theta\mathbf{d}$ are both feasible solutions. Deduce that $\mathbf{x}$ is not an extreme point solution.

(b) Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 15 \\ 12 \end{pmatrix}, \qquad \mathbf{x} = (3, 5, 2, 0)^T.$$

Find $\mathbf{d}$ and θ as in (a). Write $\mathbf{x}$ as a convex combination of two distinct feasible solutions, at least one of which is a basic feasible solution. Justify every part of your answer.

Question MTH5115–2 (Basic feasible solutions and extreme points)**.** Consider a linear program in standard equation form, maximise $\mathbf{c}^T\mathbf{x}$ subject to $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$, where A is a real $m \times 4$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{c} \in \mathbb{R}^4$. Let $\mathbf{x} = (x_1, x_2, 0, 0)^T$ be a basic feasible solution with $x_1, x_2 > 0$.

(a) (i) Prove that if $\mathbf{y}$ is a feasible solution with $y_3 = y_4 = 0$, then $\mathbf{y} = \mathbf{x}$.
(ii) Prove directly from the definitions that $\mathbf{x}$ is an extreme point solution.

(b) Now consider a linear program in standard equation form with three variables, where

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 3 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 7 \\ 5 \end{pmatrix}.$$

Find all basic feasible solutions, explaining your method.

Question MTH5115–3 (Combinations of optimal solutions)**.** Consider a linear program in standard equation form, maximise $\mathbf{c}^T\mathbf{x}$ subject to $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$, where A is a real $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{c} \in \mathbb{R}^n$.

(a) Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be feasible solutions, and suppose that $\mathbf{x} = \frac{1}{3}(\mathbf{u}+\mathbf{v}+\mathbf{w})$ is an optimal solution.
(i) Prove that $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are optimal solutions.
(ii) Let α, β, γ be real numbers (not necessarily non-negative) with $\alpha + \beta + \gamma = 1$, and suppose that $\mathbf{p} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$ satisfies $\mathbf{p} \geq \mathbf{0}$. Prove that $\mathbf{p}$ is an optimal solution.

(b) A linear program in standard equation form has three variables and objective function $c_1x_1 + c_2x_2 + c_3x_3$; its constraints are not given. You are told that $\mathbf{u} = (5, 7, 1)^T$, $\mathbf{v} = (1, 9, 3)^T$ and $\mathbf{w} = (6, 2, 2)^T$ are feasible solutions, and that $\mathbf{x} = \frac{1}{3}(\mathbf{u}+\mathbf{v}+\mathbf{w})$ is an optimal solution with optimal value 20.
(i) Find c_1, c_2 and c_3.
(ii) Find an optimal solution with at least one zero entry.

Question MTH5115–4 (Unbounded linear programs)**.** Consider a linear program in standard equation form, maximise $\mathbf{c}^T\mathbf{x}$ subject to $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$, where A is a real $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{c} \in \mathbb{R}^n$.

(a) Let $\mathbf{x}$ be a feasible solution, and let $\mathbf{d} \in \mathbb{R}^n$ satisfy $\mathbf{d} \geq \mathbf{0}$ and $A\mathbf{d} = \mathbf{0}$.
(i) Prove that if $\mathbf{c}^T\mathbf{d} > 0$, then the linear program is unbounded. (Hint: consider $\mathbf{x} + t\mathbf{d}$ for $t \geq 0$.)
(ii) Prove that if $\mathbf{d} \neq \mathbf{0}$ and $t > 0$, then $\mathbf{x} + t\mathbf{d}$ is not an extreme point solution.

(b) Consider the linear program

$$\max \; s\,x_1 - x_2 - x_3 \quad \text{subject to} \quad x_1 - x_2 = 1, \quad x_2 - x_3 = 2, \quad x_1, x_2, x_3 \geq 0,$$

where s is a real number.

(i) Find a non-zero vector $\mathbf{d} \geq \mathbf{0}$ with $A\mathbf{d} = \mathbf{0}$, and find all basic feasible solutions.

(ii) Determine the values of s for which the linear program is unbounded. For the remaining values of s, find an optimal solution and the optimal value.

Question MTH5123–1 (Existence, uniqueness and blow-up)**.** Consider equations $y' = f(x,y)$, where f and $\partial f/\partial y$ are continuous on $\mathbb{R}^2$. You may use the Picard–Lindelöf theorem.

(a) Suppose that $f(-x,-y) = f(x,y)$ for all (x,y). Prove that a solution of $y' = f(x,y)$, $y(0) = 0$ on an interval $(-h,h)$ is an odd function. Suppose instead that $f(x,y) \geq y^2$ for all (x,y). Prove that if y is a solution of $y' = f(x,y)$ on $[x_1,\beta)$ with $y(x_1) = c > 0$, then $\beta \leq x_1 + 1/c$.

(b) Consider $y' = x^2 + y^2$, $y(0) = 0$. Verify the hypotheses of the Picard–Lindelöf theorem on the rectangle $[-a,a]\times[-b,b]$, find the length $h(a,b)$ of the guaranteed interval $[-h,h]$, and find the largest possible value of $h(a,b)$. Compute the Picard iterates y_1, y_2, starting from $y_0 \equiv 0$. Show that the solution is odd and that it cannot be extended to a solution on the whole real line.

Question MTH5123–2 (Stability and the pendulum)**.** Consider an autonomous system $\dot{\mathbf{z}} = F(\mathbf{z})$ in $\mathbb{R}^2$, with F continuously differentiable, whose solutions exist for all $t \geq 0$.

(a) Let $\mathbf{p}$ be an equilibrium, and let $E\colon \mathbb{R}^2 \to \mathbb{R}$ be continuous, with $t \mapsto E(\mathbf{z}(t))$ non-increasing along every solution. Suppose that E has a strict local minimum at $\mathbf{p}$: there is $r > 0$ such that $E(\mathbf{z}) > E(\mathbf{p})$ whenever $0 < |\mathbf{z} - \mathbf{p}| \leq r$. Prove that $\mathbf{p}$ is Lyapunov stable.

(b) The damped pendulum satisfies $\ddot{\theta} + \gamma\dot{\theta} + \sin\theta = 0$, with $\gamma \geq 0$. Write it as a first-order system for $(x,y) = (\theta,\dot{\theta})$, find all equilibria, and use linearisation to classify them for $\gamma > 0$. For $\gamma = 0$, show that linearisation is inconclusive at $(0,0)$, and use (a) to prove that $(0,0)$ is stable but not asymptotically stable. Interpret your results physically.

Question MTH5123–3 (Boundary value problems)**.** (a) Let f be continuous on $[0,\pi]$. Show that $y_p(x) = \int_0^x \sin(x-t)f(t)\,dt$ satisfies $y_p'' + y_p = f$ and $y_p(0) = 0$. Prove that the boundary value problem $y'' + y = f$, $y(0) = y(\pi) = 0$ has a solution if and only if $\int_0^\pi f(x)\sin x\,dx = 0$, in which case it has infinitely many solutions. Prove that $y'' + y = f$, $y(0) = y(\pi/2) = 0$ has exactly one solution.

(b) Decide whether $y'' + y = f$, $y(0) = y(\pi) = 0$ has solutions for $f \equiv 1$ and for $f(x) = \cos x$, and find all solutions when they exist. Solve $y'' + y = 1$, $y(0) = y(\pi/2) = 0$. Explain how these examples illustrate the theorem of alternative.

Question MTH5123–4 (Exact equations and integrating factors)**.** Let P, Q be continuously differentiable on an open rectangle $D \subseteq \mathbb{R}^2$.

(a) Prove that if $\partial P/\partial y = \partial Q/\partial x$ on D, then there is a function F on D with $F_x = P$, $F_y = Q$, and that F is constant along every integral curve of $P\,dx + Q\,dy = 0$ in D. Prove that if $Q \neq 0$ on D and $(P_y - Q_x)/Q$ depends only on x, then the equation has an integrating factor $\mu = \mu(x)$ on D.

(b) Show that $(x^2 + y^2 + x)\,dx + xy\,dy = 0$ is not exact, find an integrating factor $\mu(x)$, and find the integral curves. Is the line $x = 0$ an integral curve? Find the points where both coefficients vanish. Show that two different integral curves pass through one of them, and that the other is a strict local minimum of the function F you found.

Throughout, $i > 0$ is an effective annual rate of interest, $v = (1+i)^{-1}$, and for an integer $p \geq 1$, $i^{(p)} = p\big((1+i)^{1/p} - 1\big)$ and $d^{(p)} = p\big(1 - v^{1/p}\big)$.

Question MTH5124–1 (Annuities and loan rescheduling). (a) Prove that the present value of payments of $1/p$ at the end of each p-th part of a year for n years is $a^{(p)}_{\overline{n}|} = (1-v^n)/i^{(p)}$, and that if the payments are made at the start of each period it is $\ddot{a}^{(p)}_{\overline{n}|} = (1-v^n)/d^{(p)}$. Prove that a loan L can be repaid by level annual payments X in arrears if and only if $X > iL$, and that it is then repaid after n payments, the last possibly reduced, where n is the smallest integer with $n \geq -\ln(1 - iL/X)/\ln(1+i)$.

(b) Ada repays a loan by monthly payments of £300 in advance for the next 8 years, at an APR (effective annual rate) of 4.5%. Find the outstanding loan, and the level quarterly payment in arrears over the next 12 years that repays it. Another bank offers annual payments of £1000 in arrears: show that these never repay the loan. Find how many annual payments of £1500 in arrears are needed, and the amount of the final, reduced payment.

Question MTH5124–2 (Increasing and decreasing annuities). For $n \geq 1$, let $(Ia)_{\overline{n}|} = \sum_{k=1}^{n} k\, v^k$ and $(Da)_{\overline{n}|} = \sum_{k=1}^{n} (n+1-k)\, v^k$, where $a_{\overline{n}|} = \sum_{k=1}^{n} v^k$ and $\ddot{a}_{\overline{n}|} = \sum_{k=0}^{n-1} v^k$.

(a) Prove that

$$(Ia)_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - n\, v^n}{i}, \qquad (Da)_{\overline{n}|} = \frac{n - a_{\overline{n}|}}{i},$$

and that $(Ia)_{\overline{n}|} \to (1+i)/i^2$ as $n \to \infty$.

(b) A pension pays £10,000 at the end of the first year, increasing by £500 each year, for 20 years. At $i = 4\%$, find its present value and the level annual payment in arrears for 20 years with the same present value. Find also the present value of payments of £20,000, £19,000, …, £1,000 at the ends of years $1, \ldots, 20$, and check the two values against the identity $(Ia)_{\overline{n}|} + (Da)_{\overline{n}|} = (n+1)\, a_{\overline{n}|}$.

Question MTH5124–3 (Bond pricing). A bond pays coupons at the annual rate C, in instalments of C/p at the end of each p-th part of a year, and is redeemed for R after n years. Its price at a yield i is the present value of these payments at rate i.

(a) Prove that the price is $P = C\, a^{(p)}_{\overline{n}|} + R\, v^n = R + \big(C - R\, i^{(p)}\big) a^{(p)}_{\overline{n}|}$. Deduce that the bond is priced above R if and only if $C/R > i^{(p)}$, and that P is a strictly decreasing function of i.

(b) A government bond of nominal £100 pays coupons of 3% per annum half-yearly in arrears and is redeemed at par in 10 years. Find its price at yields of 4% and of 2%, and say in each case whether it is priced at a premium or a discount. Explain what happens to the prices of existing bonds when interest rates rise.

Question MTH5124–4 (Share valuation). (a) Prove that if dividends $D(1+g)^{k-1}$ are paid at times $k = 1, 2, 3, \ldots$, where $-1 < g < i$, then their present value is $D/(i-g)$, and that the present value is infinite if $g \geq i$. Deduce that if dividends of D are paid at times $1, \ldots, m$, and the dividend then grows at rate $g < i$ per annum (so that $D(1+g)$ is paid at time $m+1$), the present value is

$$D\, a_{\overline{m}|} + \frac{D(1+g)\, v^m}{i-g}.$$

(b) A start-up is expected to pay annual dividends of 6p per share, the first in one year, constant for 3 years and then growing at 3.5% per annum. Find a fair price per share at $i = 5.5\%$.

Repeat with a growth rate of 4.5%, and comment on the sensitivity of the price to the assumed growth rate.

Unless a question says otherwise, mortality follows the Standard Ultimate Life Table (SULT) of Dickson, Hardy and Waters, with ${}_tp_x = l_{x+t}/l_x$. The workbook `SULT.xlsx` (supplied in the "Resources and Lecture Notes for Y2 modules" folder on QMPlus) contains this table, with l_x and $\ddot{a}_x$ for ages 20 to 130, and allows the interest rate (cell E4) and the parameters A, B, c of the mortality law (cells B4 to B6) to be changed. Enter interest rates unrounded, for example `=1.05/1.02-1`. Find the values of l_x and $\ddot{a}_x$ that you need in the workbook and state them in your solution, l_x to 1 decimal place and $\ddot{a}_x$ to 4 decimal places. A client with a pension fund F chooses between two options.

- *Annuity:* bought with the whole fund F as a single premium and payable annually in advance for life. It is priced on the SULT at an interest rate i, using the equivalence principle, by an insurer with initial expenses $\alpha = £1{,}500$, renewal expenses $\beta = £50$ with each payment after the first, and (where stated) a profit loading of πF, meaning that the insurer keeps a proportion π of the purchase price F as profit.
- *Drawdown:* the fund stays invested at a constant annual return r, and the same amounts as under the annuity are withdrawn annually in advance, starting now. The fund *fails* on the first withdrawal date on which it cannot pay the amount due in full. On death, the remaining fund passes to the client's estate.

The return r takes each of three given values with probability $\frac{1}{3}$, independently of the client's lifetime. A cautious, balanced or adventurous client accepts drawdown only if the probability that the fund fails while the client is still alive is at most 5%, 25% or 50%, respectively.

Question MTH5125–1 (Life annuity or drawdown)**.** A client aged 66 has $F = £300{,}000$. The annuity is a level B a year, priced at $i = 4\%$ with $\pi = 6\%$. Drawdown withdraws $W = B$ a year, with $r = 4\%$, 6% or 8%.

(a) Prove that a fund F earning $r > 0$, from which W is withdrawn at times $0, 1, 2, \ldots$, never fails if and only if $W \le dF$, where $d = r/(1+r)$, and that otherwise it pays exactly n full withdrawals and fails at time n, where $n \ge 0$ is the largest integer with $W\ddot{a}_{\overline{n}|} \le F$ (valued at r). State the equivalence principle and use it to derive a formula for B.

(b) Calculate B, and the parts of F that pay for benefits, expenses and profit. For each return, find the age at which the fund fails, if it does, and the probability that the client is alive at that age. Hence find the probability that the fund fails while the client is still alive. Advise a cautious, a balanced and an adventurous client.

Question MTH5125–2 (Inflation-linked annuity)**.** A client aged 66 has $F = £300{,}000$ and wants an income that rises with inflation, assumed to be $g = 2\%$ a year. The inflation-linked annuity pays $B_0(1+g)^t$ at time $t = 0, 1, 2, \ldots$, priced at $i = 5\%$ with $\pi = 4\%$; renewal expenses also rise, to $\beta(1+g)^t$ at time $t \ge 1$. Drawdown withdraws the same amounts, with $r = 3\%$, 6% or 9%.

(a) Prove that $\sum_{t\ge0}(1+g)^t v^t\,{}_tp_x$, with $v = (1+i)^{-1}$, equals $\ddot{a}_x$ valued at $j = \frac{1+i}{1+g} - 1$. Prove that a fund F earning $r > g$, from which $W_0(1+g)^t$ is withdrawn at times $t = 0, 1, 2, \ldots$, never fails if and only if $W_0 \le \frac{j_r}{1+j_r}F$, where $j_r = \frac{1+r}{1+g} - 1$ is the real rate, and that otherwise it pays exactly n full withdrawals and fails at time n, where $n \ge 0$ is the largest integer with $W_0\ddot{a}_{\overline{n}|} \le F$ valued at j_r. Derive a formula for B_0.

(b) Calculate B_0, and the level annual amount the same insurer would pay, with level renewal expenses of β; explain the difference. For each return, find the age at which the fund fails, if it does, and the probability that the client is alive at that age. Hence find the probability that the fund fails while the client is still alive. Advise a cautious, a balanced and an adventurous client.

Question MTH5125–3 (Longevity and the life table)**.** The SULT is built from Makeham's law $\mu_x = A + Bc^x$ with $A = 0.00022$, $B = 2.7 \times 10^{-6}$, $c = 1.124$. An insurer's own pensioners are found to follow Makeham's law with $B = 2.0 \times 10^{-6}$ and the same A, c (the *revised table*). A client aged 72 with $F = £350{,}000$ is offered a level annuity of $W = £29{,}434$ a year, priced on the SULT at 5% with $\pi = 4\%$. The insurer's expected profit is F less the expected present value at 5% of the annuity payments and expenses. No values of r are needed in this question.

(a) Prove that $p_x = \exp\{-(A + Bc^x(c-1)/\ln c)\}$. Prove that decreasing B increases every ${}_tp_x$ ($t \geq 1$), and deduce that it increases the curtate expectation of life $e_x = \sum_{t\geq 1} {}_tp_x$ and $\ddot{a}_x$. Show that $e_x = \ddot{a}_x - 1$ when $i = 0$. Derive a formula for the insurer's expected profit, and deduce that decreasing B decreases it.

(b) Use (a) and `SULT.xlsx` to find e_{72} and the insurer's expected profit, each on the SULT and on the revised table. Find the annual amount the insurer could offer on the revised table with the same expected profit as it had on the SULT. Explain who bears the cost of the revision if the client buys the annuity of $£29{,}434$, and if the client chooses drawdown instead.

Question MTH5125–4 (A couple)**.** A couple, both aged 65, with independent lifetimes, have $F = £250{,}000$ and need the same income while at least one of them is alive. The annuity is a last-survivor annuity of B a year, priced at 5% with $\pi = 4\%$; drawdown withdraws $W = B$ a year while at least one is alive, with $r = 3\%$, 5% or 8%. Here the risk limits apply to the probability that the fund fails while at least one of them is still alive. You may use that the fund pays n full withdrawals, where n is the largest integer with $W\ddot{a}_{\overline{n}|} \leq F$ (valued at r), so that it fails at time n, and never fails if $W \leq dF$, $d = r/(1+r)$.

(a) Prove that the probability that at least one of them is alive after t years is $2\,{}_tp_{65} - ({}_tp_{65})^2$, that the expected present value $\ddot{a}_{\overline{65:65}}$ of 1 a year payable in advance while at least one of them is alive equals $2\ddot{a}_{65} - \ddot{a}_{65:65}$, where $\ddot{a}_{65:65} = \sum_{t\geq 0} v^t ({}_tp_{65})^2$, and that $\ddot{a}_{\overline{65:65}} > \ddot{a}_{65}$. Derive a formula for B.

(b) Calculate $\ddot{a}_{65:65}$ by adding a column to `SULT.xlsx`. Calculate B, and the single-life annuity the insurer would pay one of them for the whole fund. For each return, find the age at which the fund fails, if it does, and the probability that at least one of them is alive then. Hence find the probability that the fund fails while at least one of them is still alive. Compare this with the corresponding probability for one partner alone who withdraws the same amount W, and advise cautious, balanced and adventurous couples.

A random variable $X \geq 0$ has the (two-parameter) *Pareto distribution* with parameters $\alpha > 0$, $\lambda > 0$ if $P(X > x) = \left(\frac{\lambda}{\lambda+x}\right)^{\alpha}$ for $x \geq 0$.

Question MTH5126–1 (Pareto losses and their excesses)**.** (a) Prove that if U is uniform on $(0,1)$, then $\lambda\big((1-U)^{-1/\alpha}-1\big)$ has the Pareto distribution with parameters α, λ. Prove that if X is Pareto with parameters α, λ and $u \geq 0$, then conditionally on $X > u$, the excess $X - u$ is Pareto with parameters α and $\lambda + u$; deduce that for $\alpha > 1$ the mean excess loss $E[X - u \mid X > u]$ is $(\lambda + u)/(\alpha - 1)$. Contrast this with an exponentially distributed loss.

(b) Let $\alpha = 3$, $\lambda = 1$. For $u = 1, 2, 3, 4$, find $P(X > u)$, the expected number of values exceeding u in a simulated sample of size $25{,}000$, the mean excess loss, and $F_u(1)$, where $F_u(x) = P(X - u \leq x \mid X > u)$. Describe how the graph of F_u on $[0, 10]$ changes with u. A simulated sample of size $25{,}000$ gave the empirical value 0.1463 for $F_1(0.1)$; is this consistent with the theory?

Question MTH5126–2 (Excess of loss reinsurance)**.** Individual claims X are Pareto with parameters $\alpha > 1$ and λ. Under an excess of loss treaty with retention $M > 0$, the insurer pays $\min(X, M)$ and the reinsurer pays $(X - M)^+ = \max(X - M, 0)$.

(a) Prove that

$$E\big[(X-M)^+\big] = \frac{\lambda^{\alpha}}{(\alpha-1)(\lambda+M)^{\alpha-1}}, \qquad E\big[\min(X,M)\big] = \frac{\lambda}{\alpha-1}\Big(1 - \Big(\frac{\lambda}{\lambda+M}\Big)^{\alpha-1}\Big).$$

Prove that if all claims are inflated by a factor $k > 1$ and M is unchanged, the reinsurer's expected payment per claim increases by a factor greater than k.

(b) Let $\alpha = 3$, $\lambda = 2000$ and $M = 3000$. Find the probability that a claim involves the reinsurer, the expected payments per claim of the insurer and of the reinsurer, and the reinsurer's expected payment on a claim that involves it. Find the effect of 10% claims inflation on both expected payments, and comment.

Question MTH5126–3 (The chain ladder method)**.** Let $C_{i,j}$ be the cumulative claims of accident year $i = 0, \ldots, n$ by development year $j = 0, \ldots, n$, known for $i + j \leq n$; claims are fully run off after n years. The chain ladder development factors are $f_j = \sum_{i=0}^{n-j} C_{i,j} \big/ \sum_{i=0}^{n-j} C_{i,j-1}$, $j = 1, \ldots, n$.

(a) Prove that f_j is a weighted average of the individual link ratios $C_{i,j}/C_{i,j-1}$, $i = 0, \ldots, n-j$, and identify the weights. Prove that if $C_{i,j} = c_i \beta_j$ for all i, j, with all $c_i, \beta_j > 0$, then the chain ladder method reproduces the ultimate claims $C_{i,n}$ exactly.

(b) An insurer's cumulative incurred claims are as follows.

Accident year	Dev. 0	Dev. 1	Dev. 2	Dev. 3	Dev. 4
2018	245	267	274	289	292
2019	255	276	288	294	
2020	265	283	292		
2021	263	278			
2022	271				

Using the basic chain ladder method (no allowance for inflation), find the reserve at the end of 2022, by accident year and in total. Verify the weighted average in (a) for development year 1. Comment on the reliability of the reserve for each accident year.

Question MTH5126–4 (The Bornhuetter–Ferguson method)**.** For accident year i, let C_i be the latest cumulative claims, $F_i \geq 1$ the product of the chain ladder development factors from the latest

development year to ultimate, P_i the premium, and ELR an initial estimate of the ultimate loss ratio, so that $U_i^0 = \mathrm{ELR}\, P_i$ is the initial estimate of the ultimate claims. The chain ladder reserve is $R_i^{CL} = C_i F_i - C_i$; the Bornhuetter–Ferguson reserve is $R_i^{BF} = (1 - 1/F_i)\, U_i^0$.

(a) Explain why, under the chain ladder assumptions, a proportion $1/F_i$ of the ultimate claims of year i is expected to have been reported, and hence justify the formula for R_i^{BF}. Prove that $R_i^{BF} - R_i^{CL} = (1 - 1/F_i)\big(U_i^0 - C_i F_i\big)$, so that the two reserves for year i agree if and only if U_i^0 equals the chain ladder ultimate. Prove that if the latest cumulative claims C_i increase by h, the chain ladder reserve increases by $h(F_i - 1)$, while the Bornhuetter–Ferguson reserve does not change.

(b) For accident years 2018–2022, the latest cumulative claims are $292, 294, 292, 278, 271$, the premiums are $306, 312, 318, 326, 330$, and the chain ladder development factors are $f_1 = 1.07393$, $f_2 = 1.03390$, $f_3 = 1.03737$, $f_4 = 1.01038$ (2018 is fully run off, 2022 is at development year 0). Taking ELR to be the ultimate loss ratio of 2018, find the Bornhuetter–Ferguson reserve for each year and in total, and compare with the chain ladder reserves. Find both reserves for 2022 if its claims at development year 0 had been 291 instead of 271, and comment on which method you would prefer for the most recent accident year.

Throughout, $X_1, X_2, \ldots$ are independent random variables with common mean μ and variance $\sigma^2 \in (0, \infty)$, $S_n = X_1 + \cdots + X_n$ and $\bar{X}_n = S_n/n$.

Question MTH5129–1 (The weak law of large numbers)**.** (a) Prove Markov's inequality, and deduce Chebyshev's inequality. Prove that $\mathrm{Var}(S_n) = n\sigma^2$, and deduce the weak law of large numbers in the form $P(|\bar{X}_n - \mu| \le \varepsilon) \ge 1 - \sigma^2/(n\varepsilon^2)$ for every $\varepsilon > 0$.

(b) A poll estimates an unknown proportion p by the proportion $\bar{X}_n$ of 'yes' answers among n independent respondents. Using (a), find the smallest n guaranteeing $P(|\bar{X}_n - p| \le 0.05) \ge 0.95$ for every p. Find the n suggested by the normal approximation, and explain the difference.

Question MTH5129–2 (The central limit theorem and rescaling)**.** Assume in addition that the X_i are identically distributed, and let $W_n = (S_n - n\mu)/n^\beta$ with $\beta > 0$. You may use the central limit theorem.

(a) Prove that if $\beta > \frac{1}{2}$, then $P(|W_n| > \varepsilon) \to 0$ for every $\varepsilon > 0$; that if $\beta = \frac{1}{2}$, then W_n/σ converges in distribution to $N(0,1)$; and that if $\beta < \frac{1}{2}$, then $P(|W_n| \le K) \to 0$ for every $K > 0$. Deduce the approximate distributions of S_n and $\bar{X}_n$ for large n.

(b) Let the X_i be Poisson with mean 1, so that S_n is Poisson with mean n. Show directly, using moment generating functions, that $(S_n - n)/\sqrt{n}$ converges in distribution to $N(0,1)$. Approximate $P(S_{100} \le 110)$ with and without a continuity correction, and compare with the exact value 0.8529.

Question MTH5129–3 (The F-test for two variances)**.** Independent random samples of sizes n_1, n_2 from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ have sample variances S_1^2, S_2^2. Let $F_{\nu_1,\nu_2;q}$ denote the q-quantile of the F_{ν_1,ν_2} distribution.

(a) Prove that $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$ has the F_{n_1-1,n_2-1} distribution, explaining the role of the χ^2 distribution. Prove that $F_{\nu_1,\nu_2;q} = 1/F_{\nu_2,\nu_1;1-q}$. Derive a $100(1-\alpha)\%$ confidence interval for σ_1^2/σ_2^2, and prove that the two-sided F-test at level α rejects $H_0\colon \sigma_1^2 = \sigma_2^2$ if and only if this interval does not contain 1.

(b) The first sample has $n_1 = 13$, mean 9.5 and variance 93.3; the second has $n_2 = 11$, mean 14.0 and variance 25.2. Given $F_{12,10;0.975} = 3.621$ and $F_{10,12;0.975} = 3.374$, test H_0 at the 5% level and find a 95% confidence interval for σ_1^2/σ_2^2. What does this imply for a comparison of the two means?

Question MTH5129–4 (The matched pairs t-test)**.** Let (A_i, B_i), $i = 1, \ldots, n$, be independent pairs of measurements, and suppose the differences $D_i = A_i - B_i$ are $N(\mu_D, \sigma_D^2)$.

(a) Prove that $T = \sqrt{n}(\bar{D} - \mu_D)/S_D$ has the t_{n-1} distribution, and derive the matched pairs t-test of $H_0\colon \mu_D = 0$ and a confidence interval for μ_D. Prove that $\sigma_D^2 = \sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B$, where ρ is the correlation of A_i and B_i, and use this to explain when the matched pairs test must be used instead of a two-sample t-test, and why it can be more powerful.

(b) Eight pairs of patients, matched by age and weight, received treatments A and B, with blood test results

A	14.0	5.0	8.6	11.6	12.1	5.3	8.9	10.3
B	13.2	4.7	9.0	11.1	12.2	4.7	8.7	9.6

Given $t_{7;0.95} = 1.895$, $t_{7;0.975} = 2.365$, $t_{14;0.975} = 2.145$, carry out the matched pairs test at the 5% and 10% levels, and find a 90% confidence interval for μ_D. Compare with the (invalid) two-sample t-test, and explain the difference using (a).

In Questions 1 and 4, $Y_1, \ldots, Y_n$ are independent Pareto random variables with known minimum $\lambda > 0$ and unknown shape $\alpha > 0$, with density $f(y;\alpha) = \alpha\lambda^\alpha y^{-\alpha-1}$ for $y > \lambda$, and $T = \sum_{i=1}^n \log(Y_i/\lambda)$. The data for these questions are $n = 10$ claim sizes (in units of $\lambda = 1$):

$$1.2,\ 1.5,\ 1.1,\ 2.4,\ 1.3,\ 3.0,\ 1.7,\ 1.05,\ 1.9,\ 1.4.$$

Question MTH5131–1 (Maximum likelihood)**.** (a) Find the maximum likelihood estimator $\hat\alpha$ of α and the Fisher information $I(\alpha)$, and hence the lower bound of Cramér and Rao for the variance of unbiased estimators of α. Prove that each $\log(Y_i/\lambda)$ is exponential with rate α, so that $T \sim \mathrm{Gamma}(n,\alpha)$, and deduce that for $n \geq 3$, $\hat\alpha$ is biased, $(n-1)/T$ is unbiased, and its variance $\alpha^2/(n-2)$ exceeds the Cramér–Rao lower bound.

(b) For the data above, compute $\hat\alpha$ and the unbiased estimate $(n-1)/T$, and state the approximate large-sample distribution of $\hat\alpha$, estimating its variance from the data. Explain why the bias of $\hat\alpha$ cannot be judged from this single value.

Question MTH5131–2 (Efficiency of unbiased estimators)**.** Let $Y_1, \ldots, Y_n$ be a random sample from the exponential distribution with rate λ, and let $\mu = 1/\lambda$. Consider $T_1 = \bar Y$ and $T_2 = nY_{(1)}$, where $Y_{(1)} = \min(Y_1, \ldots, Y_n)$.

(a) Prove that the Cramér–Rao lower bound for unbiased estimators of μ is μ^2/n, and that T_1 attains it. Prove that $Y_{(1)}$ is exponential with rate $n\lambda$, and deduce that T_2 is unbiased with variance μ^2. Prove that among all unbiased estimators of the form $\sum_i c_i Y_i$, T_1 has the smallest variance.

(b) Waiting times $2.1,\ 0.4,\ 3.5,\ 1.2,\ 0.8$ are observed. Compute both estimates of μ and their estimated standard errors. How large must n be for the standard error of T_1 to be at most 10% of μ, and what is the answer for T_2? Explain why being unbiased for every n does not make T_1 and T_2 equally good.

Question MTH5131–3 (Bias, mean square error and consistency)**.** Let $Y_1, \ldots, Y_n$ be a random sample from the uniform distribution on $(0,\theta)$, and consider $T_1 = 2\bar Y$, $T_2 = Y_{(n)} = \max(Y_1, \ldots, Y_n)$ and $T_3 = \frac{n+1}{n} T_2$.

(a) Find the bias, variance and mean square error of T_1, T_2 and T_3. Prove that an estimator whose mean square error tends to 0 is consistent, and deduce that all three estimators are consistent.

(b) Prove that for $n \geq 2$, T_3 has the smallest mean square error of the three, and determine for which n the biased estimator T_2 beats the unbiased T_1. For the sample $0.6,\ 4.8,\ 1.1,\ 0.9,\ 1.6$, compute the three estimates, and explain why the value of T_1 is not a sensible estimate of θ.

Question MTH5131–4 (Bayesian updating)**.** Suppose α has a $\mathrm{Gamma}(a,b)$ prior distribution, with density proportional to $\alpha^{a-1}e^{-b\alpha}$, where $a, b > 0$ are known.

(a) Prove that the posterior distribution of α is $\mathrm{Gamma}(n+a, T+b)$. Prove that the Bayes estimator under squared error loss is the posterior mean, and that this equals $Z\tilde\alpha + (1-Z)\alpha_0$, where $\tilde\alpha = n/T$, $\alpha_0 = a/b$ and $Z = T/(T+b)$. Prove that $Z \to 1$ with probability one as $n \to \infty$.

(b) With prior $a = 6$, $b = 2$ and the data above, find the posterior distribution, the Bayes estimate, Z, and the posterior standard deviation. Interpret the result.

Throughout, $x_1, \ldots, x_n$ are known constants and $y_1, \ldots, y_n$ are independent observations. Use $z_{0.025} = 1.96$.

Question MTH5120–1 (Least squares in matrix form)**.** Consider the regression through the origin $y_i = \beta x_i + \epsilon_i$, $i = 1, \ldots, n$, with $E[\epsilon_i] = 0$, $\mathrm{Var}(\epsilon_i) = \sigma^2$ and the ϵ_i uncorrelated, written in matrix form as $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\epsilon}$.

(a) Write down $\mathbf{Y}$, the design matrix $\mathbf{X}$, β and $\boldsymbol{\epsilon}$, with their dimensions. State the ordinary least squares (OLS) estimator of β in matrix form, show that it equals $\hat{\beta} = \sum x_i y_i / \sum x_i^2$, and explain what is required for $\mathbf{X}^T\mathbf{X}$ to be invertible. Prove that $\hat{\beta}$ is unbiased and that $\mathrm{Var}(\hat{\beta}) = \sigma^2 / \sum x_i^2$.

(b) Write down the hat matrix $\mathbf{H}$ for this model and show that it is symmetric and idempotent, and that the residual vector $\mathbf{Y} - \mathbf{H}\mathbf{Y}$ is orthogonal to the column of $\mathbf{X}$. For $x = (1, 2, 3, 4)$ and $y = (2.1, 3.9, 6.2, 7.8)$, compute $\hat{\beta}$, the fitted values and the residuals, verify the orthogonality, and estimate σ^2 and the standard error of $\hat{\beta}$.

Question MTH5120–2 (The Gauss–Markov theorem)**.** For the model of Question MTH5120–1 with all $x_i \neq 0$, an alternative estimator of the slope is the average of the individual slopes, $\tilde{\beta} = \frac{1}{n} \sum_{i=1}^{n} y_i / x_i$.

(a) Show that $\tilde{\beta}$ is a linear estimator of the form $\sum w_i y_i$, prove that it is unbiased, and derive $\mathrm{Var}(\tilde{\beta})$. State the Gauss–Markov theorem, including its assumptions.

(b) Prove the Gauss–Markov theorem for this model: every linear unbiased estimator $\sum w_i y_i$ of β has variance at least $\sigma^2 / \sum x_i^2$, with equality only for the OLS estimator $\hat{\beta}$. For $x = (1, 2, 3, 4)$ and $\sigma^2 = 1$, compute $\mathrm{Var}(\hat{\beta})$ and $\mathrm{Var}(\tilde{\beta})$, and explain why $\tilde{\beta}$ is less precise.

Question MTH5120–3 (Maximum likelihood for binomial data)**.** Suppose y_i is the number of successes in x_i independent trials, each with success probability β, so that $y_i \sim \mathrm{Bin}(x_i, \beta)$ with mean $\mu_i = \beta x_i$ and probability mass function $f(y_i; \beta) = \binom{x_i}{y_i} \beta^{y_i} (1 - \beta)^{x_i - y_i}$, $y_i = 0, 1, \ldots, x_i$.

(a) Derive the log-likelihood $\ell(\beta)$ for n independent observations. Find the maximum likelihood estimator $\hat{\beta}$ by solving the score equation, and verify that it is a maximum. Determine whether $\hat{\beta}$ is unbiased, derive the Fisher information, and show that $\mathrm{Var}(\hat{\beta})$ equals the Cramér–Rao lower bound.

(b) Construct an approximate 95% confidence interval for β, justifying the approximation. Four batches of $x = (10, 20, 30, 40)$ trials give $y = (3, 5, 9, 11)$ successes; compute $\hat{\beta}$ and the interval. Compare $\hat{\beta}$ with the average of the four batch proportions y_i / x_i, and explain which you prefer.

Question MTH5120–4 (Maximum likelihood when the variance is not constant)**.** Suppose $y_i \sim N(\beta x_i, \sigma^2 x_i)$, where $x_i > 0$ and σ^2 is known, so that both the mean and the variance grow with x_i.

(a) Derive the log-likelihood $\ell(\beta)$ for n independent observations and find the maximum likelihood estimator $\hat{\beta}$ by solving the score equation. Show that $\hat{\beta}$ is unbiased, derive the Fisher information, and show that $\mathrm{Var}(\hat{\beta})$ equals the Cramér–Rao lower bound.

(b) Show that under this model the OLS estimator $\sum x_i y_i / \sum x_i^2$ is still unbiased, with variance $\sigma^2 \sum x_i^3 / (\sum x_i^2)^2 \geq \sigma^2 / \sum x_i$. Explain why this does not contradict the Gauss–Markov theorem. For $x = (1, 2, 3, 4)$, $y = (2.1, 3.9, 6.2, 7.8)$ and $\sigma^2 = 1$, compute $\hat{\beta}$, a 95% confidence interval for β, and the variances of both estimators.